



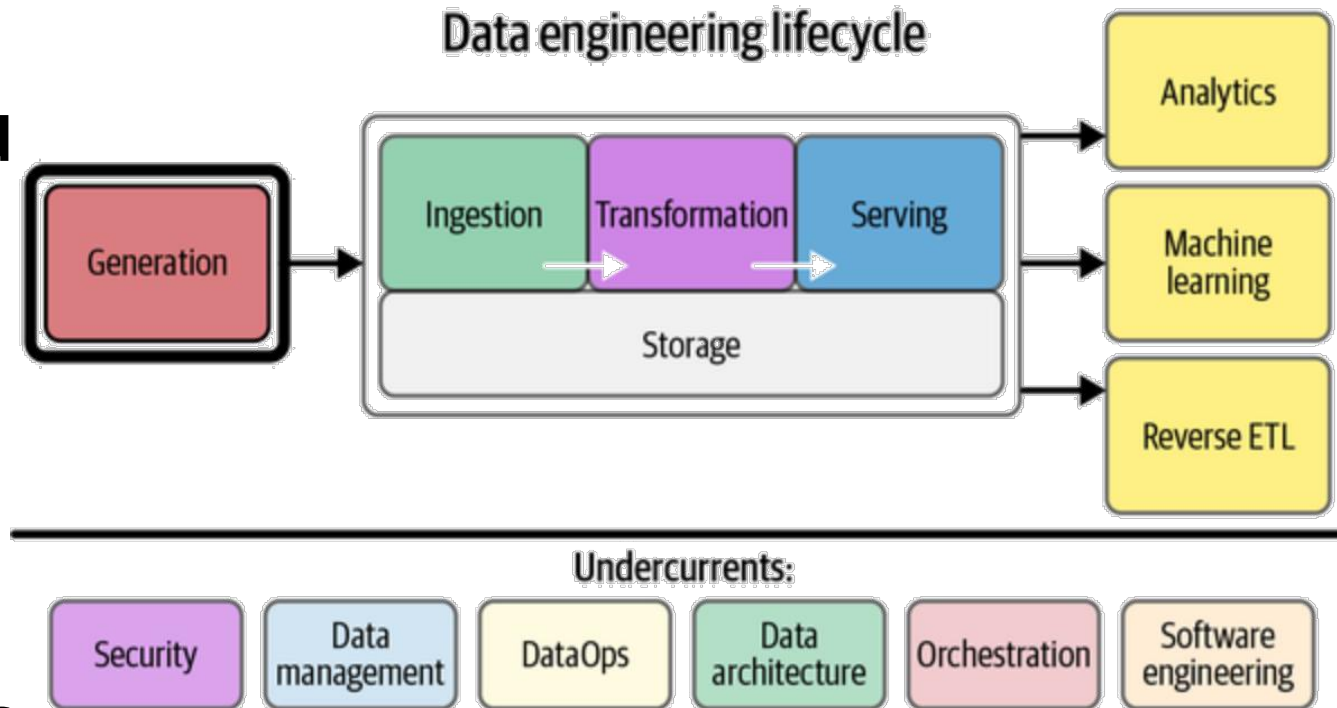
Fundamentals of Data Engineering

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Data engineering

- Data engineering is the development, implementation, and maintenance of systems and processes that take in raw data and produce high-quality, consistent information that supports downstream use cases, such as analysis and machine learning.
- Data engineer manages data engineering lifecycle, beginning with getting data from source systems & ending with serving data for use cases, such as analysis or machine learning.



Traditional ETL vs Hadoop ELT

- **ETL stands for Extract, Transform and Load.**
- **The ETL process typically extracts data from the source/transactional systems, transforms it to fit the model of data-warehouse and finally loads it to the data warehouse.**
- **The transformation process involves cleansing, enriching and applying transformations to create desired output.**
- **Data is usually dumped to a staging area after extraction.**
- **ELT stands for Extract, Load and Transform.**
- **As opposed to loading just the transformed data in the target systems, the ELT process loads the entire data into the data lake. This results in faster load times.**
- **The load process can also perform some basic validations and data cleansing rules.**
- **The data is then transformed for analytical reporting as per demand.**



Data storage

- **Data storage is related to multiple stages in data engineering life cycle i.e. ingestion, transformation and serving.**
- **Storage needs to be selected based on read/write requirement, speed, durability, consistency, availability, scalability, fault tolerance, ... factors.**
- **Storage tradeoffs**
 - **Local storage vs Distributed storage**
 - **Strong consistency vs Eventual consistency**
- **Storage options are: File storage, Local disk storage, Network attached storage (NAS), Cloud file systems (S3/Blob), Block storage, RAID, Storage area network (SAN), Object storage, HDFS, Streaming storage.**



Batch processing vs Stream processing



Batch processing

1st day of every month

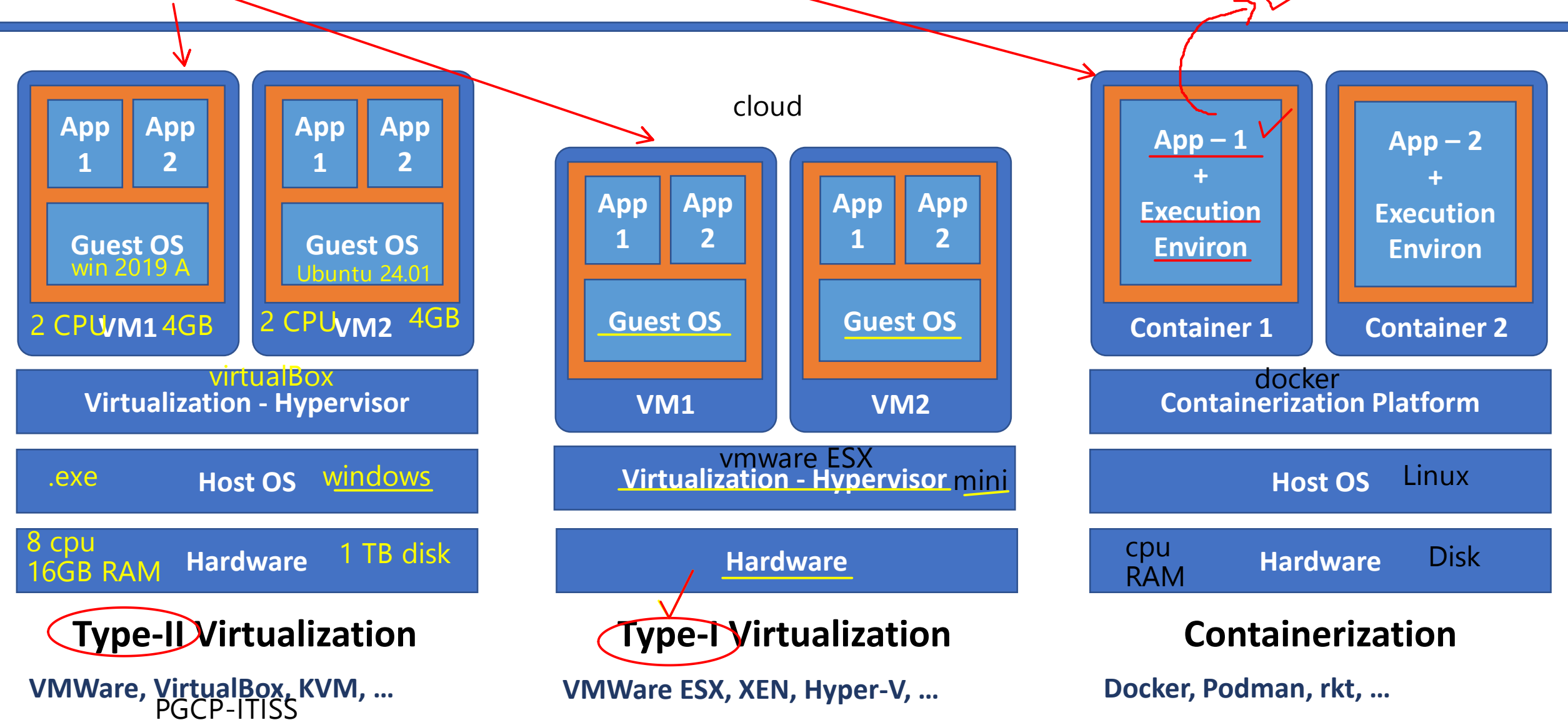
- Processing finite set of data (data at rest).
- Incremental data load is managed by programmer.
- Cluster planned as per data size.
- ✓ High throughput.
- Job run once per batch.

Stream processing

- Processing live stream of data (data in motion).
- Data processing is managed by the framework.
- Less throughput.
- Job is running forever. ✓



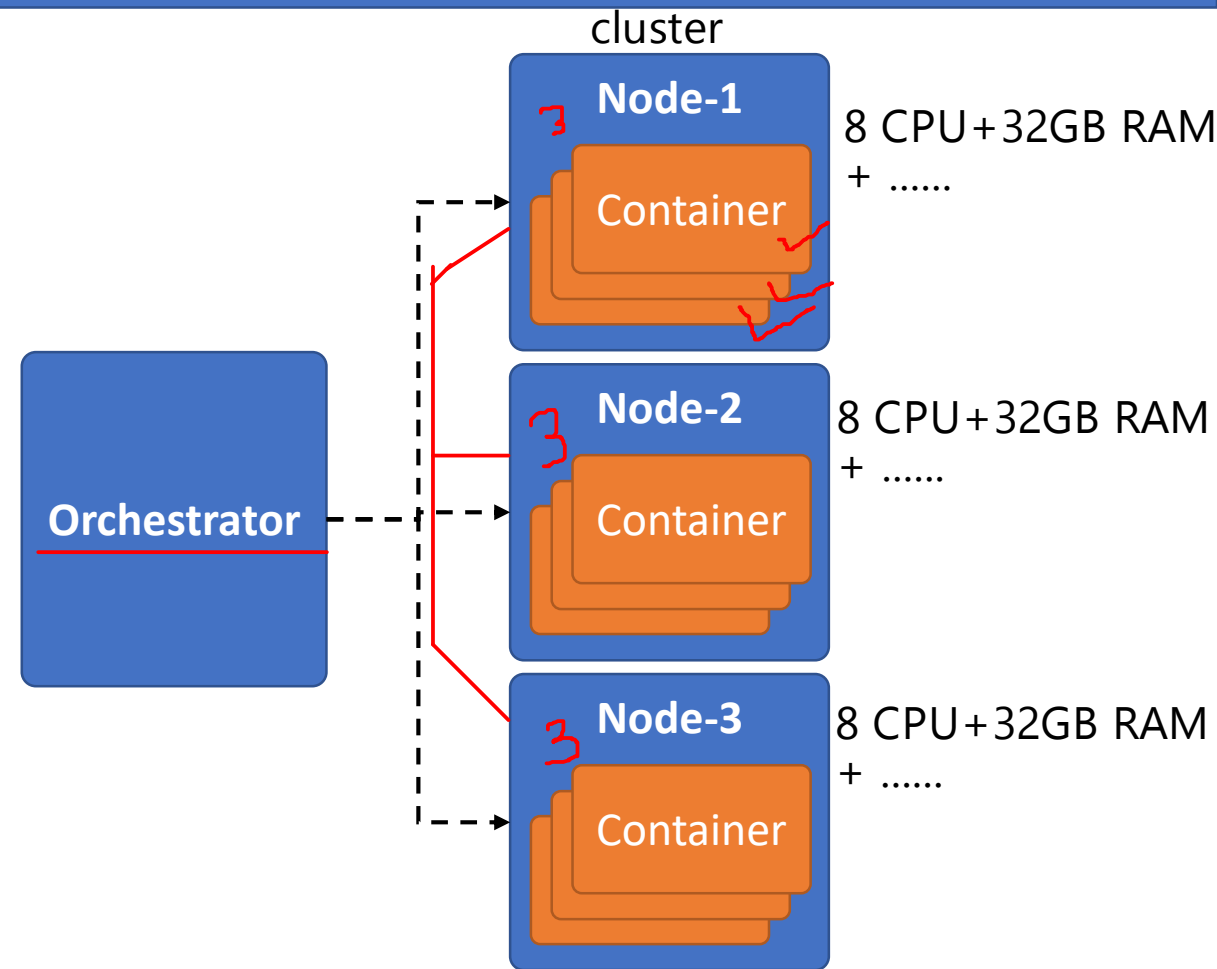
Virtualization vs Containerization



Orchestration

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- **Container Orchestration** auto increase or decrease containers to handle change in workloads/demands. It also handles container failure (re-start).
- **Ex: Docker swarm, Kubernetes, ...**

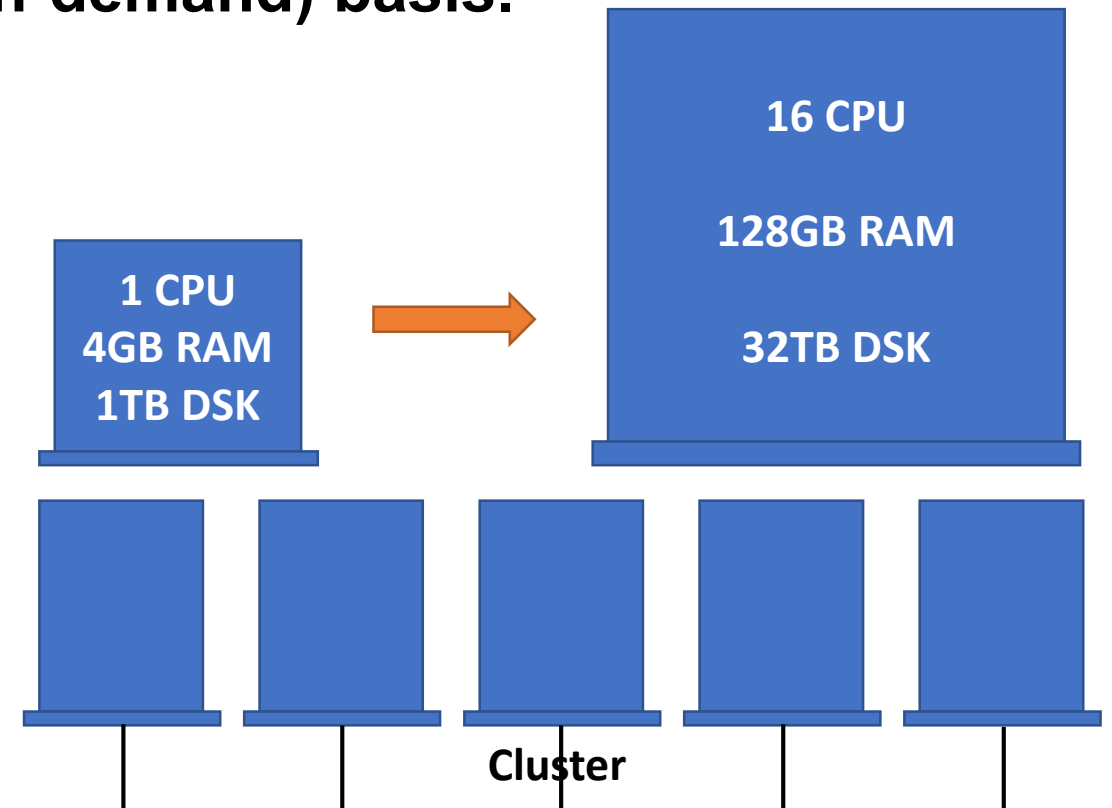


Scalability and Elasticity

- Scalability is “ability of system / application to perform well under an increased or expanding workload”.
- The resource usage is increased or decreased as per workload.
- Vertical scalability / Up scaling:
 - Increasing single system (hardware) resources in order to handle higher loads.
 - Need to handle SPOF (single point of failure) by adding backup system.
- Horizontal scalability / Out scaling:
 - Adding new systems/nodes into the cluster in order to handle higher loads.
 - More economical solution with higher complexity.

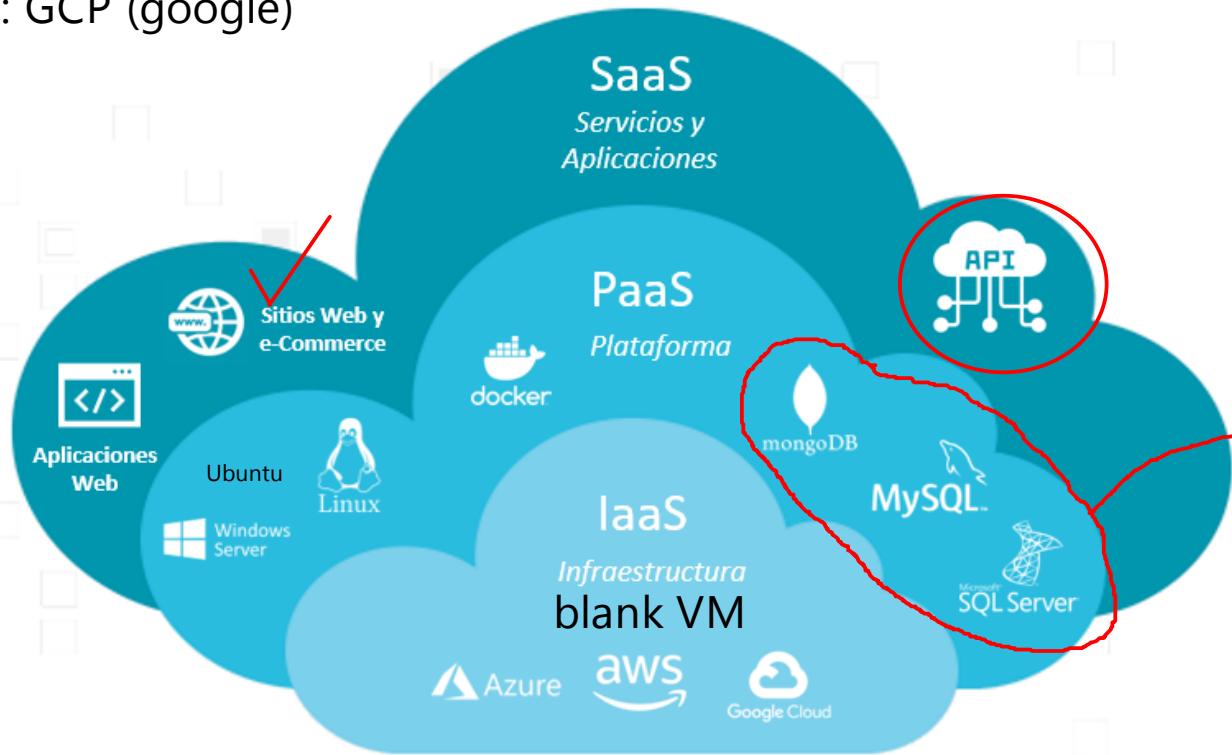
type 1

- Elastic: Cloud systems are designed to increase/decrease load as per workload.
- Cloud payments are usually pay-per-use (on-demand) basis.



Cloud Service Models

cloud providers
1: AWS (Amazon)
2: Azure (MS)
3: GCP (google)



- IaaS: Infrastructure as-a Service^{VM}
 - AWS EC2, S3, VPC
- PaaS: Platform as-a Service^{OS}
 - Beanstalk, SageMaker,
- SaaS: Software as-a Service
 - Gmail, Drive, Facebook, LinkedIn, Netflix
- DaaS: Database as-a Service
 - RDS, Aurora, Atlas, DynamoDb
- FaaS: Function as-a Service
 - Lambda, Google functions

aws



Big Data & Analytics Spectrum

- **Data storage**

- RDBMS & NoSQL databases
- Data warehouse
- S3, DFS, ...



- **Data Analysis & visualizations**

- Data Visualizations
- Business reports



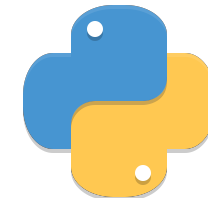
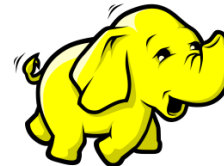
- **Artificial Intelligence, Data Science & Data mining**

- Mathematics, Statistics & Computer algorithms
- Machine learning & Deep learning
- R Programming, Python



- **(Big) Data Engineering**

- Hadoop, Hive, Spark, Kafka, BigTable, ...
- Java, Scala, Python.



- **Infrastructure**

- Linux, Cloud Computing

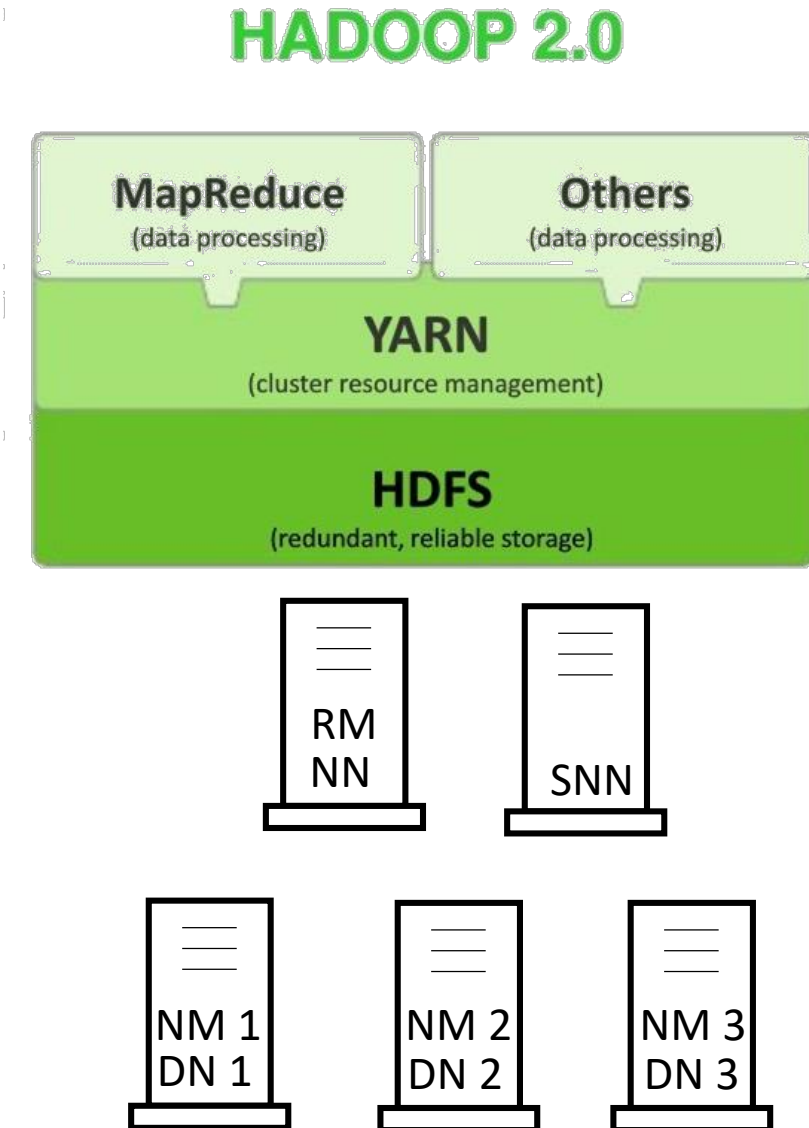
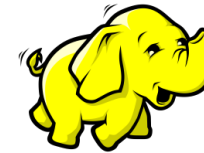


Apache Hadoop

2003 -> Google FS => DFS

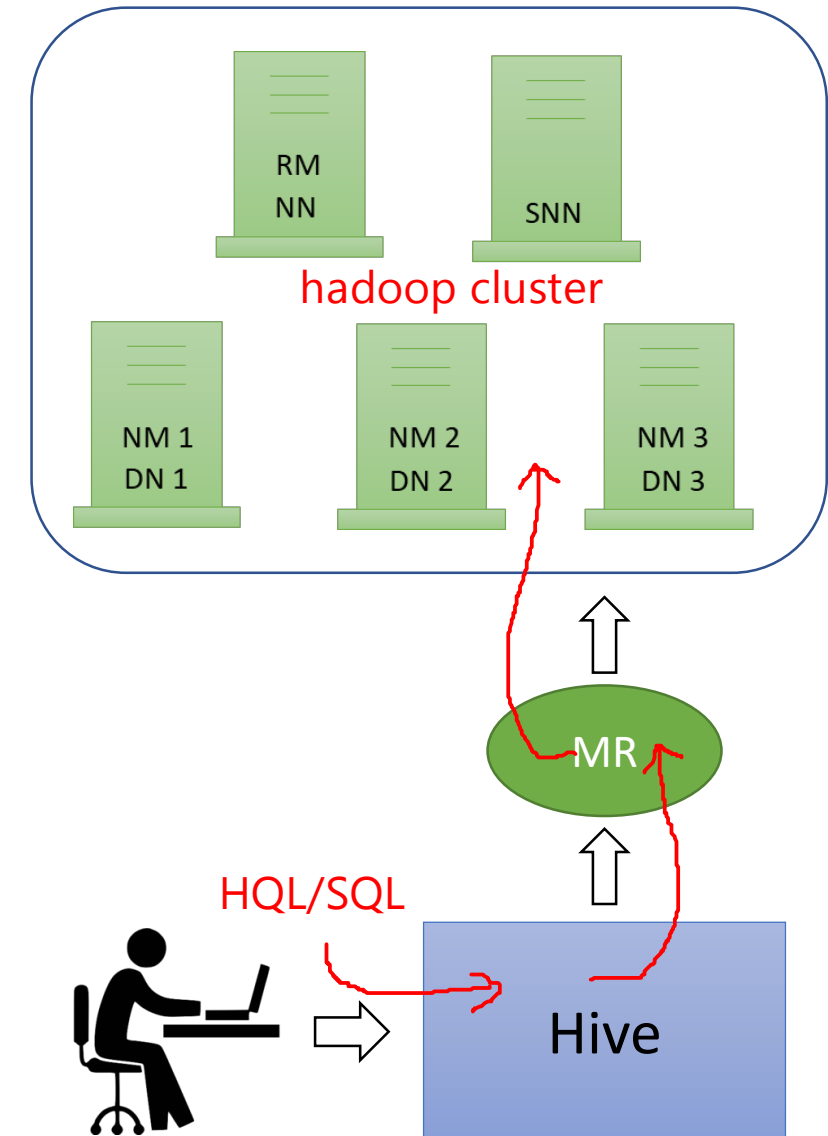
2004 -> map reduce

- Hadoop is developed by Doug cutting.
 - Web crawler – Nutch
 - Distributed computing and storage needed to process huge data produced by the crawler.
 - Joined Yahoo. Developed and open sourced under Apache license. 2006
- Hadoop
 - Distributed storage: HDFS hadoop distributed file system
 - Distributed computing Map-reduce java
 - Cluster manager: YARN yet another resource negotiator
- Hadoop is like a Kernel/Platform on which many different applications are built (eco-systems).



Apache Hive

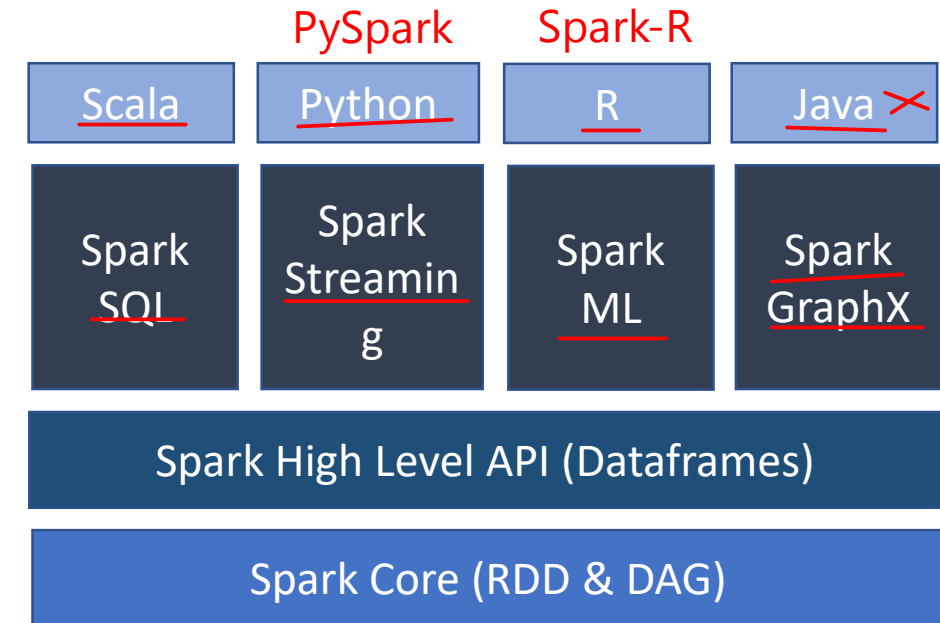
- Developed by Facebook (2007)
- Client software that convert Hive QL queries to MapReduce.
- Hive QL is similar to SQL with many extended features.
- Hive manages structured data. RDBMS
- Hive is data warehouse (OLAP) built for Hadoop.
 - Data storage = HDFS
 - Metadata = RDBMS
 - Data processing = Map-reduce/ Hive/ Spark /Tez.



Apache Spark

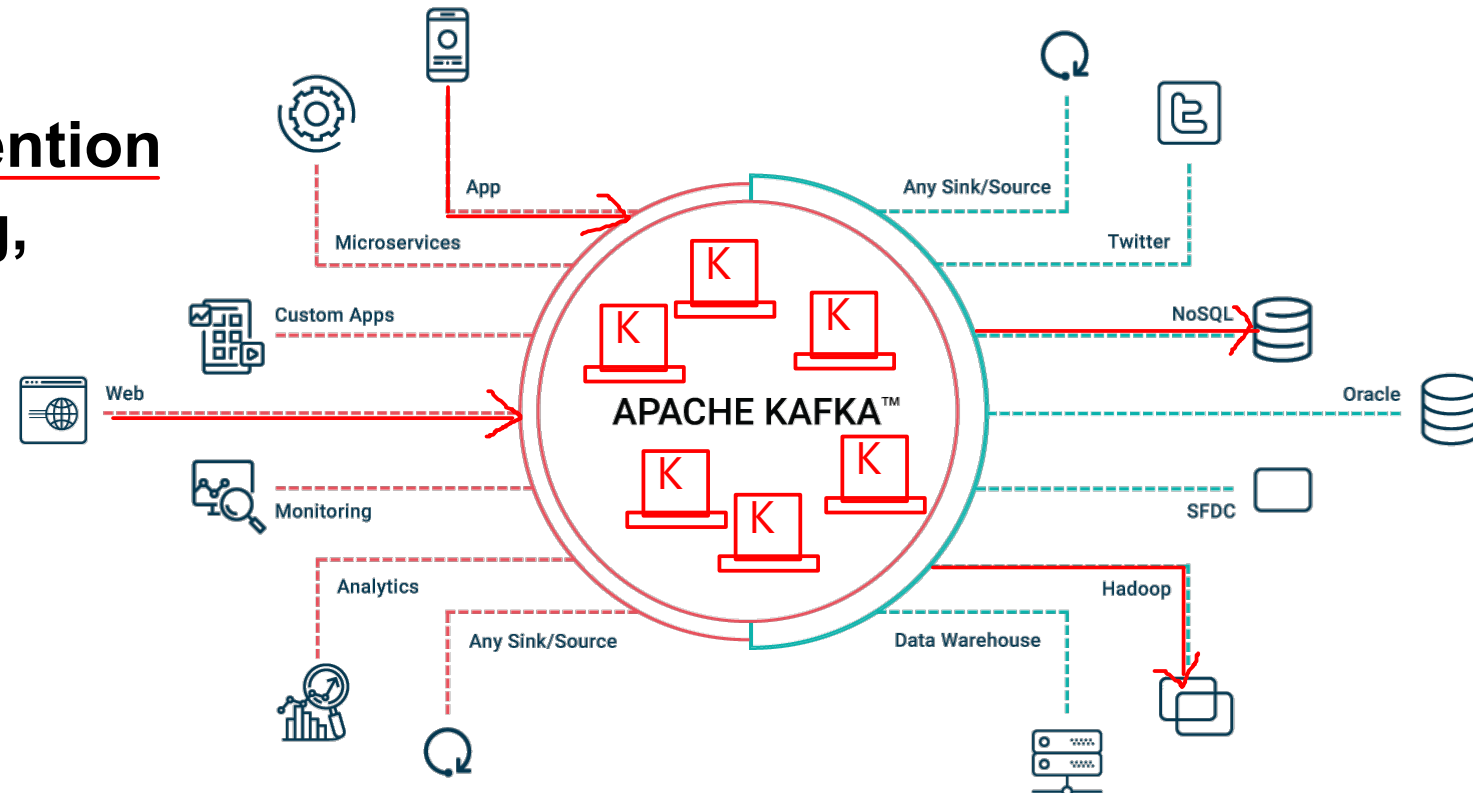
storage => HDFS

- Spark is Distributed computing framework, that can process huge amount of data.
- Spark can be used as eco-system of Hadoop or can be used as independent distributed computing framework.
- Developed by UCB AMPLabs division.
- Further developed/maintained by DataBricks.
- Popular Spark vendors
 - DataBricks, AWS EMR, Cloudera, MapR
- Spark Toolkit



Apache Kafka

- Kafka is a distributed messaging system.
- Developed at LinkedIn and open sourced in 2011.
- Used by LinkedIn, Twitter, Uber, airbnb, ...
- **Advantages**
 - Scalable, Durable, Finite retention
 - Low latency, Strong ordering,
 - Exact once delivery
- **Applications**
 - Stream processing
 - Notifications.



Big Data domains & opportunities

- Domains: Health-care, Retails, Trading/Share market, Finance, Security, Fraud, Search engines, Log Analysis, Telecom, Traffic Control, Manufacturing and lot more.
- Big Data is all about :- Think, Collect, Manage, Analyze, Summarize, Visualize, Discover Knowledge and Take Decisions.

- Job profiles:

- Business Analyst/Intelligence
- Database engineer / DWH
- Big Data engineer
- Data operations
- Big Data Architect





Thank you!

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